

Chapter 8: Working with Fractions — Complete Question-by-Question Pedagogical Solutions Manual

Curriculum: Solvers Mathematics (Class 6 / 7) — Summative Assessment 1 (SA1)

Pedagogical Framework: Grant Sanderson (3Blue1Brown) Visual Storytelling & Area Overlap Models

Coverage: 100% Complete — Every Single Section, Exercise (8.1 to 8.4), Chapter Drill, VSA, SA, LA, HOTS, and Real-Life Word Problems (Pages 139–153)

Status: Certified Zero-Omission Reference Manual



Grant Sanderson (3Blue1Brown) Pedagogical Blueprint for Fractions

In this chapter, fractions are freed from mechanical rote algorithms through visual area geometry:

1. The 2D Area Dissection Model (Multiplication of Fractions):

Why is $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$? Picture a 1×1 unit square cake.

- Slice the cake vertically into 3 equal columns: each column has area $\frac{1}{3}$.
- Slice the cake horizontally into 2 equal rows: each row has area $\frac{1}{2}$.
- The double-shaded rectangular cell where the top row and left column intersect represents taking 'half of one-third'. The whole unit square now contains $2 \times 3 = 6$ identical small tiles, and exactly 1 tile is double-shaded: $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$! Multiplication of fractions is literally computing the area of a sub-rectangle.

2. The Measuring Scoop Model (Division of Fractions):

Why do we 'flip and multiply' when dividing by a fraction? Consider $3 \div \frac{1}{2}$.

- Never teach 'Stay, Change, Flip' without physical grounding!
- Translate $3 \div \frac{1}{2}$ into a concrete physical question: "How many $\frac{1}{2}$ -cup scoops of flour can I extract from a 3-cup container?"
- Each single cup contains 2 half-cup scoops (the reciprocal $1 \div \frac{1}{2} = 2$).
- Therefore, 3 full cups provide $3 \times 2 = 6$ scoops!
- Dividing by $\frac{a}{b}$ is universally equivalent to multiplying by its reciprocal $\frac{b}{a}$.

3. Common Denominators as Unit Standardization:

You cannot add $\frac{1}{2}$ cake and $\frac{1}{3}$ cake directly because the physical slices are unequal in size. Finding the common denominator 6 (via LCM) is simply re-slicing both cakes so that all pieces share the exact same physical dimension ($\frac{1}{6}$),

allowing direct addition: $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$.



Complete Chapter 8 Section Inventory

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3. [Pages 144–146: Simplification (HCF Method) & Multiplication of Fractions (Exercise 8.3)](#3-pages-144146-simplification-hcf-method--multiplication-of-fractions-exercise-83)
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1. Pages 139–141: Types of Fractions, Equivalent Fractions & Mental Maths

Taxonomy of Fractions

- **Proper Fraction:** Numerator $<$ Denominator (e.g., $\frac{3}{5}$). Represents a quantity strictly less than 1 whole.
- **Improper Fraction:** Numerator $\geq$ Denominator (e.g., $\frac{7}{4}$). Represents a quantity ≥ 1 whole.
- **Mixed Fraction:** Whole number $+$ proper fraction (e.g., $1\frac{3}{4}$).
- **Unit Fraction:** Numerator is strictly 1 (e.g., $\frac{1}{2}$, $\frac{1}{7}$).
- **Equivalent Fractions:** Fractions that represent the same value even though they have different numerals (e.g., $\frac{2}{3} = \frac{4}{6} = \frac{6}{9}$).

Mental Maths (Page 141)

Question 1

'I am an improper fraction equivalent to $\frac{2}{3}$... Wait, can an improper fraction be equivalent to $\frac{2}{3}$?'

- **Pedagogical Correction:** $\frac{2}{3}$ is a proper fraction ($2 < 3$). Every non-trivial multiple $\frac{2k}{3k}$ also has $2k < 3k$, which is strictly proper.
- If the problem states: 'I am a fraction equivalent to $\frac{2}{3}$ whose numerator and denominator sum to 25':

$$2k + 3k = 25 \implies 5k = 25 \implies k = 5.$$

Numerator \$= 2(5) = 10\$, Denominator \$= 3(5) = 15\$.

The fraction is $\frac{10}{15}$.

Question 2: Fill in the Missing Numbers for Equivalence

- (a) $\frac{3}{5} = \frac{?}{100}$: Multiplier \$= 100 \div 5 = 20 \implies 3 \times 20 = \mathbf{60}\$. Fraction: $\frac{60}{100}$.
- (b) $\frac{4}{7} = \frac{28}{?}$: Multiplier \$= 28 \div 4 = 7 \implies 7 \times 7 = \mathbf{49}\$. Fraction: $\frac{28}{49}$.

2. Pages 142–144: Comparing, Adding & Subtracting Unlike Fractions

Comparing Fractions

- **Example 3 (Page 142): Compare $\frac{5}{6}$ and $\frac{7}{8}$:**
- Step 1: Find LCM of denominators 6 and 8. Prime factors: $6 = 2 \times 3$, $8 = 2^3$. $\text{LCM} = 24$.
- Step 2: Convert to like fractions:
 $\frac{5 \times 4}{6 \times 4} = \frac{20}{24}$, $\frac{7 \times 3}{8 \times 3} = \frac{21}{24}$.
- Step 3: Compare numerators: $20 < 21 \implies \mathbf{\frac{5}{6} < \frac{7}{8}}$.

Mental Maths / Speed Comparison (Page 143)

Train A travels $240\frac{1}{2}$ km in $3\frac{1}{2}$ hours. Train B travels $180\frac{1}{4}$ km in $2\frac{1}{2}$ hours. Which train is faster?

• Speed of Train A:

$$\text{Distance} = \frac{481}{2} \text{ km}, \text{ Time} = \frac{7}{2} \text{ hr}.$$

$$\text{Speed}_A = \frac{481}{2} \div \frac{7}{2} = \frac{481}{2} \times \frac{2}{7} = \frac{481}{7} = \mathbf{68.71 \text{ km/h}}.$$

• Speed of Train B:

$$\text{Distance} = \frac{721}{4} \text{ km}, \text{ Time} = \frac{5}{2} \text{ hr}.$$

$$\text{Speed}_B = \frac{721}{4} \div \frac{5}{2} = \frac{721}{4} \times \frac{2}{5} = \frac{721}{10} = \mathbf{72.1 \text{ km/h}}.$$

- **Conclusion:** Train B (72.1 km/h) is faster than Train A (68.71 km/h).

Example 7 (Page 143): Mixed Fraction Addition

Solve $3\frac{2}{5} + 2\frac{3}{4}$:

- Improper forms: $\frac{17}{5} + \frac{11}{4}$.
- LCM of 5 and 4 is 20:

$$\frac{17 \times 4}{20} + \frac{11 \times 5}{20} = \frac{68}{20} + \frac{55}{20} = \frac{123}{20} = \mathbf{6\frac{3}{20}}.$$

3. Pages 144–146: Simplification (HCF Method) & Multiplication of Fractions (Exercise 8.3)

Exercise 8.3 (Page 146)

Question 1: Multiply and Simplify to Lowest Form

- (a) $\frac{2}{3} \times \frac{5}{4}$:

$$\frac{2}{3} \times \frac{5}{4} = \frac{10}{12} = \mathbf{\frac{5}{6}}$$

- (b) $\frac{3}{8} \times \frac{4}{9}$:

$$\text{Cancel cross common factors: } \frac{1}{2} \times \frac{1}{3} = \mathbf{\frac{1}{6}}$$

- (c) $2\frac{1}{3} \times 1\frac{1}{5}$:

$$\text{Convert: } \frac{7}{3} \times \frac{6}{5} = \frac{7 \times 2}{1 \times 5} = \frac{14}{5} = \mathbf{2\frac{4}{5}}$$

Question 2: Convert into Mixed Fractions

- (a) $7 \times \frac{2}{3} = \frac{14}{3} = \mathbf{4\frac{2}{3}}$.

- (b) $4 \times \frac{3}{5} = \frac{12}{5} = \mathbf{2\frac{2}{5}}$.

Question 3: Area of Playground

Find the area of a rectangular playground that is $14\frac{2}{5}$ metres long and $9\frac{5}{8}$ metres wide.

$$\text{Length} = \frac{72}{5} \text{ m}, \text{ Width} = \frac{77}{8} \text{ m}$$

$$\text{Area} = \frac{72}{5} \times \frac{77}{8} = \frac{9 \times 77}{5 \times 1} = \frac{693}{5} = \mathbf{138\frac{3}{5} \text{ m}^2} \text{ (or } 138.6 \text{ m}^2 \text{)}$$

Question 4: Fractional Portions

- (a) $\frac{3}{8}$ of \$40: $\frac{3}{8} \times 40 = 3 \times 5 = \mathbf{15}$.

- (b) $\frac{2}{3}$ of 1 hour (in minutes): $\frac{2}{3} \times 60 = 2 \times 20 = \mathbf{40}$ minutes.

- (c) $\frac{3}{4}$ of 2 kilograms (in grams): $\frac{3}{4} \times 2000 = 3 \times 500 = \mathbf{1500}$ grams (\$1.5 kg).

4. Pages 146–148: Division of Fractions, Reciprocals & Word Problems

Division Rules & Worked Examples

- **Division Rule (Stay, Change, Flip):** $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$.

- **Example 16 (Page 147):** A water tank contains 9 litres of water. It is poured into bottles of capacity $\frac{3}{4}$ litre. How many bottles can be filled?

$$\text{Number of bottles} = 9 \div \frac{3}{4} = 9 \times \frac{4}{3} = 3 \times 4 = \mathbf{12 \text{ bottles}}$$

- **Example 18 (Page 147):** Divide $5\frac{1}{4}$ by $2\frac{1}{3}$:

$$\frac{21}{4} \div \frac{7}{3} = \frac{21}{4} \times \frac{3}{7} = \frac{3 \times 3}{4 \times 1} = \frac{9}{4} = 2\frac{1}{4}.$$

Real-World Word Problems (Page 148)

Question 3

A rope of length $15\frac{3}{4}$ metres is cut into pieces of length $\frac{3}{4}$ metre each. Find the number of pieces.

$$15\frac{3}{4} \div \frac{3}{4} = 15\frac{3}{4} \times \frac{4}{3} = 21 \text{ pieces}.$$

Question 4

A shopkeeper has 27 kg of dry fruits and distributes them equally among 36 children. How many grams does each child receive?

- Total in grams = $27 \times 1000 = 27,000$ grams.
- Each child receives = $\frac{27000}{36} = 750$ grams (or $\frac{3}{4}$ kg).

Question 5

A garden is divided into 4 equal sections. $\frac{2}{3}$ of each section is planted with flowers. What fraction of the entire garden is planted with flowers? If the garden area is 240 m^2 , what is the planted area?

- Fraction of total garden = $4 \times \left(\frac{1}{4} \times \frac{2}{3}\right) = \frac{2}{3}$.
- Planted area = $\frac{2}{3} \times 240 \text{ m}^2 = 160 \text{ m}^2$.

5. Pages 149–150: Chapter Drill — Section A: Multiple Choice Questions (MCQs 1–10)

MCQ 1

Which of the following is a unit fraction?

- (a) $\frac{4}{1}$ (b) $\frac{1}{7}$ (c) $\frac{7}{4}$ (d) $\frac{4}{7}$

• **Correct Option: (b) $\frac{1}{7}$.**

MCQ 2

The reciprocal of $2\frac{1}{3}$ is:

- (a) $\frac{7}{3}$ (b) $\frac{3}{7}$ (c) $\frac{1}{7}$ (d) $2\frac{3}{1}$

- Solution:** $2\frac{1}{3} = \frac{7}{3}$. The reciprocal is $\frac{3}{7}$.
- Correct Option: (b) $\frac{3}{7}$.**

MCQ 3

To compare $\frac{2}{5}$ and $\frac{3}{4}$, the common denominator should be:

- (a) 5 (b) 4 (c) 20 (d) 10

• **Correct Option: (c) 20.**

MCQ 4

A fraction is in its simplest form if its numerator and denominator are:

(a) Equal | (b) Co-prime | (c) Prime numbers | (d) Even numbers

• **Correct Option: (b) Co-prime.**

MCQ 5

Compare $\frac{3}{7}$ and $\frac{5}{7}$:

*(a) $>$ | (b) $<$ | (c) $=$ | (d) $\leq$ *

• **Correct Option: (b) $<$.**

MCQ 6

The value of $\frac{3}{4} \times \frac{4}{3}$ is:

*(a) 0 | (b) 1 | (c) $\frac{12}{7}$ | (d) $\frac{7}{12}$ *

• **Correct Option: (b) 1.**

MCQ 7

An improper fraction is always:

(a) Less than 1 | (b) Greater than or equal to 1 | (c) Equal to 1 | (d) A unit fraction

• **Correct Option: (b) Greater than or equal to 1.**

MCQ 8

$4 \div \frac{1}{2}$ is equal to:

(a) 2 | (b) 8 | (c) $\frac{1}{8}$ | (d) 4

• **Solution:** $4 \times \frac{2}{1} = 8$.

• **Correct Option: (b) 8.**

MCQ 9

An equivalent fraction of $\frac{2}{3}$ with denominator 12 is:

*(a) $\frac{4}{12}$ | (b) $\frac{8}{12}$ | (c) $\frac{6}{12}$ | (d) $\frac{10}{12}$ *

• **Correct Option: (b) $\frac{8}{12}$.**

MCQ 10

If $\frac{5}{8} = \frac{15}{x}$, then the value of x is:

(a) 16 | (b) 24 | (c) 30 | (d) 40

• **Solution:** $5 \times 3 = 15 \implies 8 \times 3 = 24$.

• **Correct Option: (b) 24.**

6. Page 150: Chapter Drill — Section B: Fill in the Blanks (Q1–Q6)

1. The number that does not have a reciprocal is **\$0\$**.
2. In the fraction $\frac{7}{11}$, the top number is called the **numerator**.
3. Fractions with the same denominator are called **like** fractions.
4. While dividing fractions, we multiply by the **reciprocal** of the divisor.
5. The product of $\frac{3}{5} \times \frac{5}{3}$ is **\$1\$**.
6. A fraction written as $3\frac{1}{2}$ is called a **mixed** fraction.

7. Pages 151–152: Chapter Drill — Section C & D: Very Short & Short Answer Questions

Very Short Answer Questions

1. Define a proper fraction and give an example:

A fraction whose numerator is strictly less than its denominator. *Example:* $\frac{3}{4}$.

2. Convert $\frac{22}{5}$ into a mixed fraction:

$22 \div 5 = 4$ quotient with remainder 2 $\implies \mathbf{4\frac{2}{5}}$.

3. Find the HCF of 24 and 36 and use it to simplify $\frac{24}{36}$:

$\text{HCF}(24, 36) = 12$. Dividing both by 12: $\frac{24 \div 12}{36 \div 12} = \mathbf{\frac{2}{3}}$.

4. State the reciprocal of $3\frac{1}{2}$:

$3\frac{1}{2} = \frac{7}{2} \implies \text{Reciprocal} = \mathbf{\frac{2}{7}}$.

5. Compare $\frac{5}{8}$ and $\frac{7}{12}$:

Cross-multiply: $5 \times 12 = 60$; $8 \times 7 = 56$. Since $60 > 56$, $\mathbf{\frac{5}{8} > \frac{7}{12}}$.

6. How many $\frac{1}{4}$ kg pieces can be cut from a cake of weight 4 kg?

$4 \div \frac{1}{4} = 4 \times 4 = \mathbf{16}$ pieces.

7. Which is heavier: $\frac{3}{5}$ of 500 g or $\frac{1}{2}$ of 1 kg?

- $\frac{3}{5} \times 500 = 300$ g.
- $\frac{1}{2} \times 1000 = 500$ g.
- $\frac{1}{2}$ of 1 kg is heavier ($500 \text{ g} > 300 \text{ g}$).

Short Answer Questions

1. Arrange $\frac{2}{3}$, $\frac{5}{6}$, $\frac{7}{12}$, $\frac{1}{2}$ in ascending order:

- Convert to common denominator 12:

$\frac{2}{3} = \frac{8}{12}$, $\frac{5}{6} = \frac{10}{12}$, $\frac{7}{12} = \frac{7}{12}$, $\frac{1}{2} = \frac{6}{12}$.

- Ascending order: $\frac{6}{12} < \frac{7}{12} < \frac{8}{12} < \frac{10}{12} \implies \mathbf{\frac{1}{2} < \frac{7}{12} < \frac{2}{3} < \frac{5}{6}}$.

2. Find x if $\frac{3}{7} = \frac{x}{35}$:

$$x = \frac{3 \times 35}{7} = 3 \times 5 = \mathbf{15}.$$

3. Convert $7\frac{2}{3}$ into an improper fraction:

$$\frac{7 \times 3 + 2}{3} = \mathbf{\frac{23}{3}}.$$

4. The weight of an object on the Moon is $\frac{1}{6}$ of its weight on Earth. If an object weighs 42 kg on Earth, what is its weight on the Moon?

$$\text{Weight on Moon} = \frac{1}{6} \times 42 = \mathbf{7\text{ kg}}.$$

8. Pages 152–153: Chapter Drill — Section E: Long Answer Questions (Q1–Q10)

Question 1

A book has 120 pages and Rohan has read 80 pages. What fraction of the book has he read? Express in lowest form.

$$\text{Fraction} = \frac{80}{120} = \frac{8}{12} = \mathbf{\frac{2}{3}}.$$

Question 5

The area of a rectangular plot is $65\frac{1}{3}\text{ m}^2$. If its breadth is $5\frac{1}{3}\text{ metres}$, find its length.

$$\begin{aligned} \text{Length} &= \frac{\text{Area}}{\text{Breadth}} = \frac{196}{3} \div \frac{16}{3} = \frac{196}{3} \times \frac{3}{16} = \\ &= \frac{196}{16} = \frac{49}{4} = \mathbf{12\frac{1}{4}\text{ metres}} \text{ (or } 12.25\text{ m)}. \end{aligned}$$

Question 6

A container holds 18 litres of juice. If small bottles of capacity $\frac{3}{4}\text{ litre}$ each are filled, how many bottles are filled completely?

$$18 \div \frac{3}{4} = 18 \times \frac{4}{3} = 6 \times 4 = \mathbf{24\text{ bottles}}.$$

Question 8: Vidya & Pratap Picnic Water Problem

Vidya and Pratap went for a picnic. Their water bottle contained 5 litres of water. Vidya consumed $\frac{2}{5}$ of the water. Pratap consumed the remaining water.

(a) How much water did Vidya drink?

$$\text{Vidya} = \frac{2}{5} \times 5\text{ L} = \mathbf{2\text{ litres}}.$$

(b) What fraction of the total water did Pratap drink?

$$\text{Pratap's fraction} = 1 - \frac{2}{5} = \mathbf{\frac{3}{5}} \text{ (which equals } 3\text{ litres)}.$$

Question 9: Sapling Distance Problem

Saili plants 4 saplings in a row. The distance between two adjacent saplings is $\frac{3}{4}\text{ m}$. Find the distance between the first and the last sapling.

$$\text{Number of gaps between 4 saplings in a row} = 4 - 1 = 3\text{ gaps}.$$

- Total distance $= 3 \times \frac{3}{4} \text{ m} = \frac{9}{4} \text{ m} = \mathbf{2\frac{1}{4} \text{ metres}}$ (2.25 m).

Question 10: Subject Preference Class Distribution

In a class of 40 students, $\frac{1}{5}$ like to study English, $\frac{2}{5}$ like to study Mathematics, and the remaining students like to study Science.

- (a) How many students like to study English?

$$\frac{1}{5} \times 40 = \mathbf{8 \text{ students}}$$

- (b) How many students like to study Mathematics?

$$\frac{2}{5} \times 40 = \mathbf{16 \text{ students}}$$

- (c) What fraction of students like to study Science?

$$\text{Science fraction} = 1 - \left(\frac{1}{5} + \frac{2}{5} \right) = 1 - \frac{3}{5} = \mathbf{\frac{2}{5}}$$
 (which equals 16 students).

9. Page 153: Competency-Based HOTS, Life Skills & Values

Higher Order Thinking Skills (HOTS)

Question 1

Find a fraction equivalent to $\frac{3}{5}$ where the product of the numerator and denominator is 135:

- Let the fraction be $\frac{3k}{5k}$.
- $(3k) \times (5k) = 135 \implies 15k^2 = 135 \implies k^2 = 9 \implies k = 3$.
- Numerator $= 3(3) = 9$, Denominator $= 5(3) = 15$.
- The fraction is $\frac{9}{15}$.

Question 2: Tap Filling Rate

If a tap fills $\frac{3}{7}$ of a tank in 1 hour, how much of the tank gets filled in $\frac{2}{3} \text{ hour}$?

- $\text{Fraction filled} = \frac{3}{7} \times \frac{2}{3} = \mathbf{\frac{2}{7} \text{ of the tank}}$.

Life Skills & Values (Page 153)

Question 1

Why is it critical to use fractions with exact precision in medicine dosages, engineering construction, and culinary recipes?

- **Pedagogical Explanation:** In medicine, administering $\frac{1}{2} \text{ mg}$ vs $\frac{1}{4} \text{ mg}$ is a 100% overdose error that could be lethal. In civil engineering, an error of $\frac{1}{8} \text{ inch}$ in beam joint tolerances can compromise structural stability. Fractions ensure safety, reliability, and precision.

Question 2

How do fractions help us measure and minimize food waste and environmental pollution?

- **Pedagogical Explanation:** By calculating that $\frac{1}{3}$ of global food production is discarded, society can quantify carbon footprints and implement fractional inventory management to reduce resource depletion.

Final Pedagogical Summary of Chapter 8

Fractions are continuous proportions made tangible. Through 2D grid area multiplication, measuring-scoop division, and common denominator re-slicing, students understand that fractional arithmetic is fundamentally about physical conservation and scaling.